

REINFORCEMENT LEARNING

LAB ASSIGNMENT - 3

SUBJECT CODE: 24DS743

Submitted by

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IN

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Task 1: Formulating the GridWorld Environment

RL Component	Description
States	S1, S2, S3, S4, S5 and the goal state G.
Actions	Up, Down, Left and Right.
Reward Structure	-1 for a normal movement and +10 for reaching the goal.
Terminal State(s)	G (Goal).
Goal of the Agent	Reach the goal while getting the highest possible cumulative reward.

GridWorld Layout

S1	S2	S3
S4	S5	G

Task 2: Policy Evaluation

The given policy was evaluated for five iterations using a discount factor of 0.9. The maximum change became zero by the fourth iteration, so the values had converged.

Iteration	Maximum Change (Δ)	Convergence Status
1	10.0000	Not Converged
2	9.0000	Not Converged
3	8.1000	Not Converged
4	0.0000	Converged
5	0.0000	Converged

State values under the evaluated policy: S1 = 6.2000, S2 = 8.0000, S3 = 10.0000, S4 = 8.0000, S5 = 10.0000, G = 0.0000.

Task 3: Value Iteration and Optimal Policy

Value Iteration converged after 4 iterations.

State	Optimal Action	State Value
S1	Down	6.2000
S2	Down	8.0000
S3	Down	10.0000
S4	Right	8.0000
S5	Right	10.0000

Optimal policy: S1 → Down, S2 → Down, S3 → Down, S4 → Right, S5 → Right.

Task 4: Optimal Path Analysis

Policy Type	Path Length	Total Reward	Goal Reached
Random Policy	20	-20	No
Evaluated Policy	3	8	Yes
Optimal Policy	3	8	Yes

Paths obtained:

Random Policy: $S1 \rightarrow S1 \rightarrow S1 \rightarrow \dots \rightarrow S1$ (20 steps); the agent never reached the goal.

Evaluated Policy: $S1 \rightarrow S2 \rightarrow S3 \rightarrow G$ (3 steps, total reward 8).

Optimal Policy: $S1 \rightarrow S4 \rightarrow S5 \rightarrow G$ (3 steps, total reward 8).

Analysis Questions

1. What is a state in GridWorld?

A state is the current position of the agent in the grid. For example, S1 represents the top-left position.

2. What actions can the agent perform?

The agent can move Up, Down, Left or Right. If it tries to move outside the grid, it stays where it is.

3. What is the purpose of Policy Evaluation?

It tells us how good a particular policy is by calculating the expected value of each state when that policy is followed.

4. How does Value Iteration differ from Policy Evaluation?

Policy Evaluation starts with a fixed policy and calculates its values. Value Iteration considers the available actions and finds the best action for each state.

5. What is the Bellman Optimality Equation?

$V^*(s) = \max_a [R(s,a) + \gamma V^*(s')]$. It means the value of a state is based on the action that gives the highest expected return.

6. Which policy reached the goal using the fewest steps?

Both the evaluated and optimal policies reached the goal in 3 steps in this experiment. The random policy did not reach the goal within 20 steps.

7. Why is the optimal policy superior to a random policy?

The optimal policy uses the state values to choose actions that lead toward the goal, while a random policy has no information about which move is useful.

8. How would obstacles affect the optimal path?

Obstacles would block some movements, so the agent may need to take a different route. This can increase the number of steps and change the optimal actions.